Voyage economics and the time charter equivalent

What a dry bulk or tanker voyage earns a day, the freight it needs, the costs that decide it, and the physical fixture against the FFA, on the Baltic Exchange's routes in 2026

Research note by Nathan Couturier, 29 October 2026. Interactive site: https://nathancouturier.github.io/voyage-tce-model/. Code and data: https://github.com/nathancouturier/voyage-tce-model.

Summary

- A VLCC fixed on TD22, Galveston to Ningbo, at the Baltic's USD 58.22 million in the week to 23 October 2026, earns about USD 466,378 a day over a 110.5 day round voyage; freight of USD 17.24 a tonne would only cover the voyage costs priced. The Baltic's own round voyage TCE that week was USD 451,905; the engine reads 3.2 percent higher. The port charges it leaves out explain only part of that: across the weeks checked the gap grows with freight.
- On routes outside Europe, bunkers are every sourced dollar of voyage cost at the August 2026 market; on voyages into or out of the EEA, EU allowances at the 2026 phase in are 12.6 to 17.4 percent of it (EUA EUR 82.31 a tonne). A Hormuz war risk premium of 7.5 percent of a USD 157 million VLCC hull would cost the charterer USD 11.78 million for each seven days inside the Gulf, 6.5 times TD3C's bunker bill.
- Round the Cape of Good Hope an LR2 on TC20 sails 31.5 more days than by Suez and needs USD 12.31 million (USD 136.77 a tonne) more freight to earn what it earns by Suez, against a Red Sea premium of USD 9.44 a tonne for its charterer; Suez stays the better way for the owner until its toll passes USD 7.43 million.
- For a VLCC from the Gulf a 7.5 percent premium costs the charterer USD 43.61 a tonne for one seven day period; 20 days of waiting inside the Gulf make 4 periods, USD 174.44 a tonne, and the owner needs USD 99.86 a tonne more freight to keep the TCE of the week to 23 October 2026.
- In the week to 3 July 2026 a C3 cargo at USD 30.11 a tonne earned USD 35,429 a day net of its commission, USD 37,294 grossed up to the gross hire the 5TC is assessed in: USD 419 a day above the 5TC FFA over its days; a C5 cargo earned USD 10,109 a day above it. By 23 October 2026 the C3 voyage sat USD 6,799 a day below the curve.
- Higher freight raises the speed that maximises the TCE and dearer bunkers lower it, on every one of the routes tested (C5, C3, TD22, TC20).

A TCE here is an estimate of what a voyage earns at stated inputs, not a forecast, and it is set against operating and capital costs, never net of them.

Of USD 58.22 million of freight, USD 51.56 million is left after commission and USD 4.48 million of voyage costs priced, all of them bunkers; port charges, with no public figure, are left out. Over 110.5 days (54.9 in ballast, 50.7 laden and 5.0 in port), the result is a TCE of USD 466,378 a day.

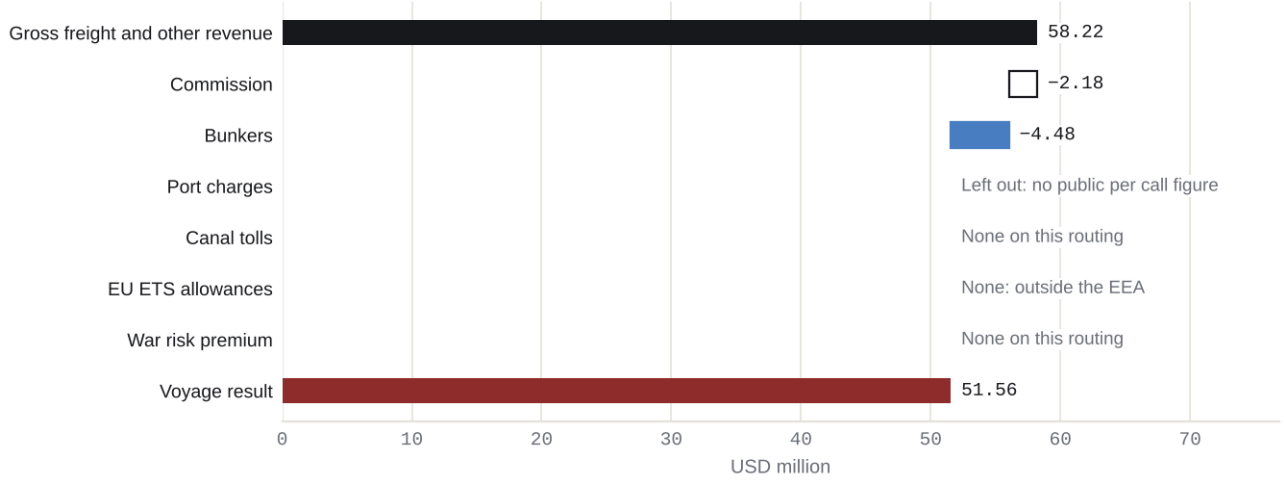


Figure 1. Engine output for TD22 in the week to 23 October 2026. Freight: Baltic Exchange via Hellenic Shipping News. Bunkers: Star Asia Shipbroking weekly market report, week 43 of 2026, 23 October 2026. A cost with no public figure is left out and named in its row. Sources retrieved 29 October 2026.

1. The question and the method

For a given ship, route and market: what does the voyage earn a day, what freight does it need, which costs decide the answer, and is fixing the cargo better than hedging it on paper? The routes are the Baltic Exchange's (Guide to Market Benchmarks v8.8, September 2026): C3, C5, P7 and the S4A trip for dry bulk; TD3C, TD22, TD20 and TD25 dirty;

TC2, TC14, TC1 and TC5 clean; TC20 and TD34 where needed.

The TCE is gross freight and other revenue, less commission and voyage costs, over the days of the whole round voyage, the ballast leg included: TCE = (freight + other revenue − commission − voyage costs) / voyage days. Voyage costs are bunkers by sulphur zone (MGO inside emission control areas, measured on the route line), port charges, canal tolls, EU allowances and, where the owner pays it, the war risk premium. The tanker TCE follows the Baltic's own method (Appendix 3 of the guide): a 5 percent weather margin on steaming days, the round voyage from the discharge port, the port and waiting days and consumptions of the route's description, and bunkers at the hub nearest the load port; C3 alone is priced at Singapore, the hub on its voyage, since the hub nearest Tubarao lies off it. Worldscale freight is WS / 100 x flat rate; flat rates are licensed and none is public for 2025 or 2026, so a Worldscale rate is priced only with the reader's own. Commission is 5 percent on dry routes and 3.75 percent on tankers, address commission and brokerage together.

Every input is sourced (a dated public document, with the passage read), derived (a stated rule), the study's choice (bulk carrier speeds: the fastest pair printed), left out as the Baltic leaves it out (a war risk premium, usually the charterer's) or empty: a figure with no public source, such as port charges and canal tolls, stays out of the TCE and the result names it. A dated market reads no price from after its date; the August 2026 common market is a period average. The engine is a Python package, mirrored in the browser and held to it within one part in a billion on 283 cases.

2. Validation

The Baltic publishes its tanker routes on a Worldscale and a TCE basis. Only two study routes are assessed as a lump sum in dollars, TD22 and TC20, so only they can be priced without a licensed flat rate: 13 weeks between 12 December 2025 and 23 October 2026. With bunker prices first printed in the week, the engine reads 3.2 to 5.2 percent above the Baltic's TCE, and 2.4 to 5.3 percent with prices up to a week old, always above it. The gap grows with freight, from USD 4,867 a day at a USD 13.68 million lump sum (23 January 2026) to USD 15,387 at USD 79.61 million (9 October 2026), so the port charges the engine leaves out, a fixed sum a voyage, explain only part of it. The weeks to 13 March 2026 and 20 March 2026, just after the closure of Hormuz, read 18.1 and 10.0 percent above with tables of the same weeks, and why is not established.

The engine reads 2.4 to 5.3 percent above the Baltic's TCE in the weeks priced with bunker prices up to a week old, except the weeks to 13 March 2026 and 20 March 2026 (18.1 and 10.0 percent above); the week to 17 April 2026, priced with Houston prices first printed on 20 March 2026, reads 0.6 percent below.

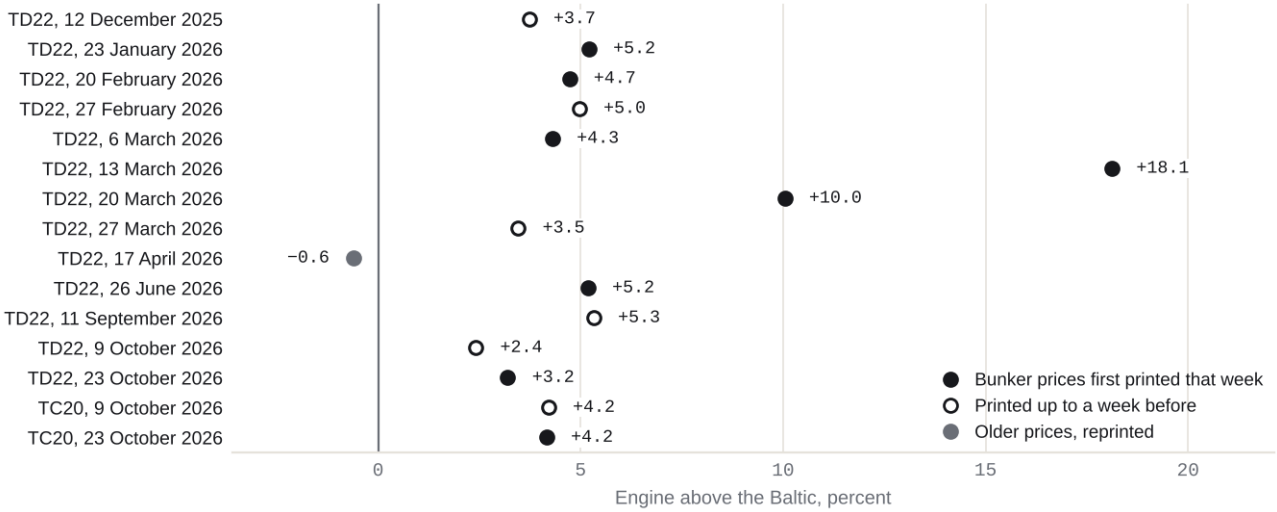


Figure 2. Baltic Exchange weekly tanker reports, weeks to 12 December 2025 to 23 October 2026, as republished by Hellenic Shipping News; bunkers: Star Asia Shipbroking and Banchero Costa weekly tables, a reprinted row dated by its first printing; engine output. Sources retrieved 29 October 2026.

The bunker table matters too: Star Asia sometimes reprints a week's table (the week to 17 April 2026, priced with Houston prices first printed on 20 March 2026, reads 0.6 percent below), and for 6 March 2026 Intermodal's Houston VLSFO, USD 689.30, against Star Asia's USD 584 moves the comparison from 4.3 to 1.7 percent. The 64 Worldscale anchors need a licensed flat rate and are recorded, not tuned. The Baltic's worked TD3C calculation of July 2017 is reproduced line by line: laden 19.85 days, ballast 20.64, 46.49 in all, 1,389.20 t of fuel laden. Searoute distances fall within 2 percent of NGA Pub. 151 for 20 of 25 port pairs; the largest miss is Port Hedland to Busan (+4.2 percent).

Early March 2026: Platts assessed the Gulf to China VLCC rate at USD 62.07 a tonne on 2 March 2026, which the engine turns into USD 319,741 a day with the bunkers of the week of 27 February 2026; with the early March war risk premium of 2.5 percent of a USD 120 million hull for one seven day period in the owner's costs, as if the freight carried it, USD 255,921. The USD 89.00 a tonne widely reported on 3 March 2026 names no primary assessment; it would give USD 468,621 a day. The Baltic's TD3C TCE was USD 209,550 in the week to 27 February 2026 and USD 485,959 in the week to 6 March 2026.

3. Where the money goes

At one common dated market, August 2026 (the EU allowance at EUR 82.31 a tonne and 1.1594 dollars a euro), sourced voyage costs are bunkers and, on voyages that touch the EEA, EU allowances. The share of MGO in the bunker bill follows the miles inside emission control areas and the days at EU berths.

Table 1. Sourced voyage costs at the August 2026 market, by route.

Route	Voyage	Days	Sourced voyage costs, USD	Bunkers, %	EU ETS, %	MGO in bunkers, %
C3	Tubarao to Qingdao	76.5	2,746,360	100.0	0.0	0.0
C5	West Australia to Qingdao	29.3	881,274	100.0	0.0	0.0
P7	US Gulf to Qingdao grain	87.0	1,507,418	100.0	0.0	8.2
S4A	US Gulf trip to Skaw-Passero	42.6	806,116	84.6	15.4	35.6
TD3C	Middle East Gulf to China	47.0	1,816,997	100.0	0.0	0.0
TD22	US Gulf to China	110.5	3,945,099	100.0	0.0	10.4
TD20	West Africa to UK-Continent	36.3	1,150,605	82.6	17.4	31.0
TD25	US Gulf to UK-Continent	40.0	1,268,237	84.5	15.5	48.1
TC2	Continent to US Atlantic coast	28.3	605,358	87.4	12.6	69.9
TC14	US Gulf to Continent	40.7	814,501	84.7	15.3	46.4
TC1	Middle East Gulf to Japan (LR2)	51.4	1,221,620	100.0	0.0	0.0
TC5	Middle East Gulf to Japan (LR1)	51.9	1,100,833	100.0	0.0	0.0

Engine output; bunkers: Ship & Bunker's August 2026 averages where it publishes them (Singapore VLSFO and HSFO; Fujairah VLSFO) and its third quarter averages, July to September, for the other hub prices (Singapore MGO; Fujairah HSFO and MGO; Rotterdam VLSFO, HSFO and MGO; Houston VLSFO, HSFO and MGO), at each route's hub; allowances: DEHSt (German auctions, proxy), monthly average for August 2026, at the 2026 phase in; dollars: Federal Reserve H.10, mean of the 21 published days of August 2026. Port charges and canal tolls have no public figure and are not in the costs. S4A is a trip time charter: its row holds the charterer's trip costs and the owner's ballast leg. Sources retrieved 29 October 2026.

4. Speed

The TCE was computed on a 0.01 knot grid from 1 knot below the slowest published consumption to 1 knot above the fastest, at the lowest, median and highest public freight of 2026 and at the lowest weekly, the August 2026 and the highest weekly bunker price at the route's hub. The textbook result holds on every route tested: freight pushes the best speed up, bunkers push it down. Tanker consumption comes at two speeds, so the fitted exponent is low and the best tanker speed sits at the top of the range.

Table 2. Laden speed that maximises the TCE, knots.

Route	Low freight	Median freight	High freight	Cheap bunkers	Dear bunkers	Fitted exponent
C5	12.0	12.0	13.0	15.0 (edge)	10.5	2.43
C3	10.7	12.0	13.0	15.0 (edge)	10.0 (edge)	2.43
TD22	13.9	14.0 (edge)	14.0 (edge)	14.0 (edge)	14.0 (edge)	2.18
TC20	14.0 (edge)	14.0 (edge)	14.0 (edge)	14.0 (edge)	14.0 (edge)	1.60

Freight columns at the common market's bunker price, bunker columns at the median freight. Edge: the best speed is at an end of the range searched. Engine output on the Baltic's published speed and consumption pairs (Guide to Market Benchmarks v8.8).

5. Detours: Suez, the Cape and Hormuz

Asia to Europe. An LR2 on TC20, Jubail to Rotterdam at the Baltic's lump sum of USD 21.51 million in the week to 23 October 2026, earns the owner USD 350,394 a day by Suez before any toll and USD 210,173 round the Cape, 31.5 days and 0.43 billion tonne miles more; the war risk premiums are the charterer's, the Red Sea's USD 9.44 a tonne on the Suez routing. A Panamax delivered at Qingdao and redelivered at Rotterdam, one way like the P4 trip, needs USD

29.01 a tonne by Suez to earn the P4 hire of USD 13,556 a day net of its commission, USD 34.54 by Panama and USD 36.08 round the Cape, before port time, port charges and tolls; at the Cape's freight, Suez could bear USD 443,218 of tolls and waiting, Panama USD 96,657.

Hormuz. A TD3C voyage spends 6.5 days inside the Gulf, one seven day period of premium; each further week of waiting there adds a period for the charterer and the TCE it forgoes for the owner. Loading off Fujairah splits the voyage in two: the export VLCC sails 2.7 fewer days, but the shuttle through the strait was reported at USD 16 to 20 a barrel (about USD 118.52 to USD 148.15 a tonne) and at USD 50 million a transit, and the transfer's own cost has no public figure. The Baltic's TD34, a cargo loading outside the strait, was assessed at USD 656,587 a day against TD3C's USD 1,287,851 in the week to 23 October 2026, freight of USD 114.38 a tonne against USD 241.39.

Per tonne of cargo the Cape costs an LR2's owner far more freight than the Red Sea premium costs its charterer, and for a VLCC loading in the Gulf each week of waiting adds a period of premium to the TCE the wait forgoes.

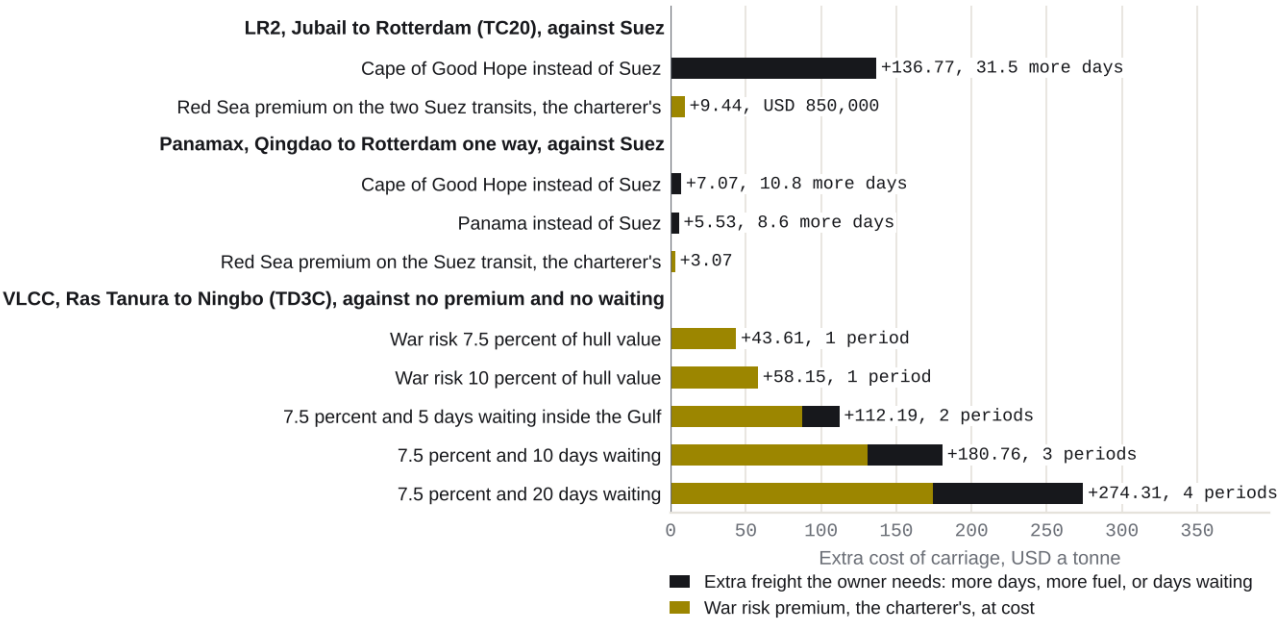


Figure 3. Engine output, week to 23 October 2026: TC20 at the Baltic's lump sum, the Panamax one way at the freight that earns the P4 hire net of its commission, TD3C at the Baltic's TCE of the week (USD 1,287,851 a day). War risk: Marsh, 22 July 2026, on Xclusiv Shipbrokers' hull values of 7 September 2026. Suez and Panama tolls and the crew war bonus have no public figure for these ships and are left out; the transfer chain and TD34 are set out in the Hormuz panel. Sources retrieved 29 October 2026.

6. Carbon

EU allowances cover half the emissions of a voyage into or out of the EEA and all of one between two EEA ports and within a port of call, for 40 percent of 2024 emissions, 70 percent of 2025 and 100 percent from 2026, when methane and nitrous oxide join carbon dioxide (Directive 2003/87/EC, Articles 3ga and 3gb and Annex I). Under the BIMCO ETS Allowances Clause for time charter parties the charterer provides and pays for the allowances for the ship's emissions during the charter, and the owner, as the shipping company, surrenders them (Article 3gc): on the S4A trip the charterer pays USD 66,690 and the owner USD 57,491 for the ballast leg back. FuelEU Maritime's penalty is assessed on a ship's yearly energy, not a voyage's, so it is shown beside the TCE and kept out of it.

Table 3. EU ETS cost of one voyage by phase in year, at the August 2026 allowance price.

Voyage	2024	2025	2026	2026, USD a tonne of cargo
TC23, between two EEA ports	USD 11,310	USD 19,793	USD 28,708	0.96
TD20, into the EEA	USD 78,791	USD 137,885	USD 200,059	1.54
TD3C, outside the EEA	USD 0	USD 0	USD 0	0.00

Engine output at EUR 82.31 a tonne (DEHSt (German auctions, proxy), monthly average for August 2026) and 1.1594 dollars a euro (Federal Reserve H.10, mean of the 21 published days of August 2026); emission factors of Regulation (EU) 2015/757 Annex I as amended, global warming potentials 28 for methane and 265 for nitrous oxide. Sources retrieved 29 October 2026.

7. Physical against paper

Capesize 5TC FFAs settle on the month's average of the daily index in dollars a day; C3, C5 and the ICE tanker routes on the month's average in dollars a tonne. Time charter assessments are gross hire, so a voyage's net TCE is grossed up

by its commission before it is set against them (guide, section 7.4.3), and the FFA is the strip over the voyage's days from the fixture date. The C3 voyage of the week to 3 July 2026 is hedged with 76.5 days of 5TC (72.7 hedge the owner's net earnings exactly) or 170,000 tonnes of C3. That day's spot 5TC, USD 33,678, is the 180,000 t basis, as the BCI and BDI of the day show: the 182,000 t ship is set against USD 37,181. A C5 cargo fixed on 22 October 2026 at USD 14.15 a tonne for 5 to 7 November 2026, against a C5 index of USD 14.10, earns USD 276 a day more than one at the index. No public tanker FFA quote in dollars a tonne was found.

Grossed up by their commission, the C5 voyages earn well above the 5TC FFA over their days, while C3 sat near it in July and USD 6,799 a day below it in October.

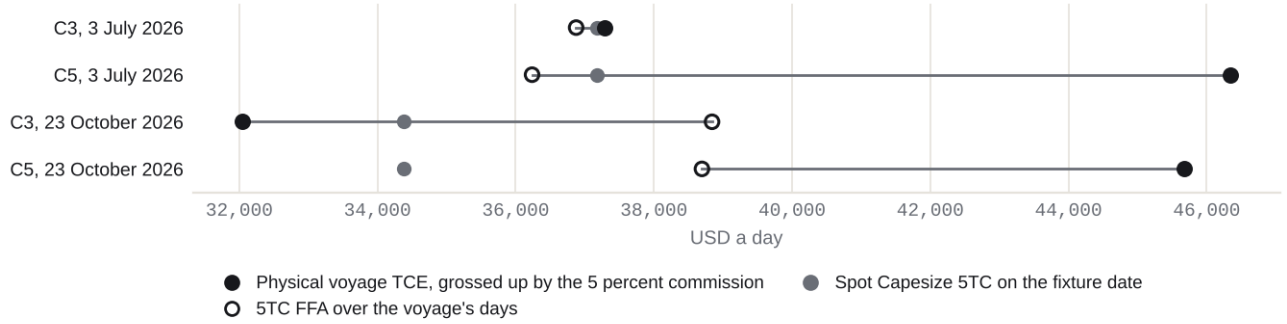


Figure 4. Engine output for C3 and C5 at the route rates and bunker tables of the fixture dates, bunkers at Singapore; FFA: Banchero Costa, Dry bulk FFA assessments, curves of 29 June 2026 and 25 October 2026; spot 5TC: Signal Group and Banchero Costa, the July figure on the 180,000 t basis and here on the 182,000 t. Sources retrieved 29 October 2026.

8. What the study cannot do

- Port charges have no public per call figure: every TCE leaves them out (each USD 100,000 is USD 1,307 a day on C3, USD 3,419 on C5 and USD 905 on TD22), and they explain only part of the gap to the Baltic, which grows with freight.
- Worldscale flat rates are licensed and none is public for 2025 or 2026: a Worldscale rate is priced only with the reader's own, and those anchors are not reproduced.
- Suez and Panama tolls, and the time at a canal on the dry routings, are left out; the detours report what each routing could bear instead.
- Bunker prices are broker tables and averages at four hubs, not what a ship paid: two brokers can differ widely for the same day, and Star Asia sometimes reprints a week's table.
- War risk premiums are the latest dated rates found, from July 2026, and the listed area is taken to end at the strait, so the count of seven day periods is a lower bound.
- The dry FFA curves are two dated assessments whose vessel basis and dates are partly unstated; no public tanker FFA quote in dollars a tonne was found.
- Bulk carrier speeds are the study's choice, the fastest pair the guide prints; tanker consumption comes as two pairs, so the best speed often sits at the top of the range searched.

Sources and reproduction

Baltic Exchange, Guide to Market Benchmarks v8.8 (September 2026) and its TD3C TCE calculation of July 2017; the Baltic's weekly tanker reports, December 2025 to October 2026, and the brokers' and insurers' reports cited above, as republished by Hellenic Shipping News; Ship & Bunker; the European Commission and DEHSt auction reports; the Federal Reserve's H.10; EUR-Lex and IMO texts; NGA Pub. 150 and Pub. 151; searoute 1.6.0 over Eurostat SeaRoute; EEX and ICE contract specifications. Every figure's document, retrieval time and hash are in the repository's data/raw/manifest.json; licensed documents are cited, never redistributed. scripts/note.py writes this note from the committed data.